

Operational Analysis Report

Unaccompanied Children (UAC) Care Transition Pipeline Analysis & Framework

1. Introduction

The Unaccompanied Children (UAC) program is responsible for the temporary care and placement of children who enter the United States without a legal guardian. After apprehension by U.S. Customs and Border Protection (CBP), children are transferred to the Department of Health and Human Services (HHS), where they receive medical care, shelter, education, and case management before being reunified with an approved sponsor.

The effectiveness of this system depends not only on available capacity but also on the speed and reliability of transitions between each stage of care. Delays in transfers or sponsor placement can increase operational costs, extend the duration of care, and reduce overall system efficiency.

This project models the UAC process as a care transition pipeline, enabling operational performance to be measured using data-driven metrics rather than relying solely on occupancy statistics.

2. Problem Statement

Current reporting practices primarily monitor the number of children in CBP custody and HHS care. While these figures indicate system utilization, they provide limited visibility into operational performance.

Key operational questions that remain unanswered include:

- How efficiently are children transferred from CBP to HHS?
- Are discharge rates sufficient to keep pace with new arrivals?
- During which periods do care backlogs accumulate?
- Is sponsor placement becoming more or less efficient over time?
- Which stages of the care pipeline contribute most to operational delays?

Without structured transition analytics, identifying bottlenecks and implementing targeted improvements becomes significantly more difficult.

3. Project Objectives

Primary Objectives

- Measure the efficiency of transitions from CBP custody to HHS care.
- Evaluate discharge performance and sponsor placement outcomes.

- Identify operational bottlenecks across the care pipeline.
- Monitor backlog accumulation and unresolved cases.
- Analyze trends in transition performance over time.

Secondary Objectives

- Improve visibility into case management workflows.
- Support faster reunification of children with approved sponsors.
- Enable evidence-based operational and policy decisions.
- Develop an interactive dashboard for continuous monitoring.

4. Dataset Description

The project utilizes daily operational records published by the U.S. Department of Health and Human Services. Each observation represents system activity for a specific reporting date.

Column	Description
Date	Reporting date
CBP Intake	Children apprehended and placed into CBP custody
CBP Custody	Active number of children in CBP custody
Transferred to HHS	Children transferred from CBP to HHS
HHS Care	Children currently receiving HHS care
Discharged	Children successfully placed with approved sponsors

These variables collectively represent the movement of children through the care transition pipeline.

5. Technology Stack

The project was developed using widely adopted open-source technologies suitable for analytical applications.

- **Python:** Core programming language
- **Pandas:** Data cleaning and transformation
- **NumPy:** Numerical computations
- **Plotly:** Interactive visualizations
- **Streamlit:** Dashboard development
- **Jupyter Notebook:** Exploratory Data Analysis

- **Git & GitHub:** Version control and project hosting

6. Methodology

The project followed a structured analytical workflow comprising the following stages:

Data Preparation

- Imported the dataset using Pandas.
- Converted the Date column into datetime format.
- Standardized column names.
- Converted numerical values from text to numeric data types.
- Removed invalid and missing records.
- Validated data consistency.

Feature Engineering

Several analytical metrics were derived to evaluate operational efficiency:

Transfer Efficiency (%)

$$\text{Transfer Efficiency} = \frac{\text{Transferred to HHS}}{\text{CBP Custody}} \times 100$$

Measures the proportion of children transferred from CBP custody to HHS care.

Discharge Efficiency (%)

$$\text{Discharge Efficiency} = \frac{\text{Discharged}}{\text{HHS Care}} \times 100$$

Measures how effectively HHS care results in successful sponsor placement.

Pipeline Throughput (%)

$$\text{Pipeline Throughput} = \frac{\text{Discharged}}{\text{CBP Intake}} \times 100$$

Measures overall pipeline performance.

Placement Ratio (%)

$$\text{Placement Ratio} = \frac{\text{Discharged}}{\text{Transferred to HHS}} \times 100$$

Measures sponsor placement relative to successful transfers.

Daily Backlog

$$\text{Daily Backlog} = \text{CBP Intake} - \text{Discharged}$$

Cumulative Backlog

Running sum of all previous daily backlogs

Unresolved Cases

$$\text{Unresolved Cases} = \text{CBP Custody} + \text{HHS Care}$$

7. Dashboard Overview

An interactive Streamlit dashboard was developed to visualize and monitor the complete care transition pipeline.

Core Modules

- Care Pipeline Flow Visualization
- Transfer Efficiency Dashboard
- Discharge Efficiency Dashboard
- Bottleneck Detection Charts
- Outcome Trend Analysis

User Features

- Date Range Selection
- Ratio-Based Metric Toggle
- Threshold-Based Visual Alerts
- Downloadable Filtered Dataset

These features enable stakeholders to monitor operational performance in real time and identify areas requiring intervention.

8. Expected Business Value

The proposed analytical framework transforms operational data into actionable performance indicators. Rather than focusing solely on custody counts, decision-makers gain visibility into transition speed, placement effectiveness, and backlog dynamics.

Potential benefits include:

- Faster identification of operational bottlenecks.

- Improved coordination between CBP and HHS.
- Enhanced monitoring of sponsor placement outcomes.
- Better allocation of shelter and case management resources.
- Data-driven support for policy evaluation and operational planning.

9. Recommendations

- Establish automated monitoring of transfer and discharge efficiency.
- Deploy threshold-based alerts for backlog growth.
- Increase sponsor verification resources during high-intake periods.
- Integrate predictive analytics to forecast future care demand.
- Expand data collection to include facility-level processing times and regional characteristics.
- Conduct periodic reviews of pipeline performance using standardized KPIs.

10. Conclusion

This project demonstrates how process-oriented analytics can provide a more comprehensive assessment of the UAC care transition system than aggregate occupancy reporting alone. By modeling the operational workflow as a structured pipeline, the analysis introduces meaningful performance indicators that support continuous monitoring, bottleneck detection, and informed decision-making.

The combination of Python-based data analysis and a Streamlit dashboard offers a scalable and user-friendly solution for operational stakeholders. While the report does not present dataset-specific findings, the methodology provides a robust framework that can be applied to real operational data to improve efficiency, transparency, and reunification outcomes.

References

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